

Increase in interest rates: what impacts on the european interbank market?

Linear and nonlinear econometrics

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Introduction

- ➔ Investigate how the normalization of the European Central Bank's (ECB) monetary policy influences the European interbank market between october 2019 at december 2023 (daily).
- ➔ Model the linear and nonlinear transmission mechanisms of monetary policy shocks on interbank transaction volumes.

Introduction

- Studies like Vari (2016) and Allen et McAleer (2020) highlight the asymmetric transmission of monetary shocks and their impact on market stability, particularly in fragmented interbank markets.
- Dynamic models such as ARDL and NARDL (Shin et al., 2014) effectively capture long-term asymmetries, while works like Schorderet (2003) refine the understanding of asymmetric cointegration.
- Bennani et al. (2017) emphasize macroprudential measures to mitigate monetary shocks, advocating for stress testing and enhanced oversight.
- Address key questions:
 - ➡ How do changes in the refinancing rate propagate through the interbank market?
 - ➡ Are the effects symmetric, or do positive and negative rate changes impact differently?
 - ➡ What are the implications for the stability of the banking system?

Context of monetary policy

The interbank market plays a central role in liquidity management and financial stability.

→ **Post-2008 crisis Zero Lower Bound pression :**

- ECB adopted historically low interest rates to counteract deflationary pressures.
- Refinancing rate remained at 0% for an extended period to support lending and economic recovery.

→ **Normalization since 2022 :**

- Rapid rate hikes by the ECB in response to rising inflation, increasing from 0% to 4.5%.
- Shift marked the end of ultra-accommodative monetary policies.

→ **Macroprudential challenges :**

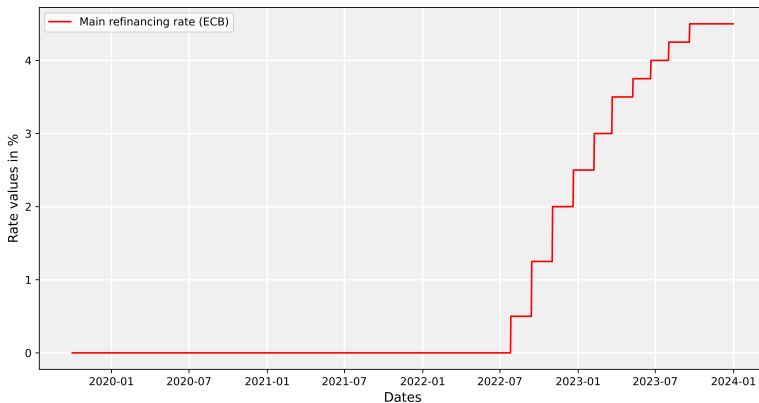
- Rate hikes increased financing costs and reduced liquidity in the interbank market.
- Amplified volatility in interbank transactions, impacting overall financial stability.

Economic Framework: ECB Key Rates

- **Main refinancing rate (REFI)**, the central instrument of the ECB's monetary policy:
 - Defines the cost of short-term liquidity for banks.
 - Banks borrow funds against collateral during open market operations.
- **Deposit facility rate**, acts as the lower boundary of money market rates:
 - Banks earn interest on excess reserves deposited with the ECB.
 - Encourages interbank lending by maintaining rates below the REFI rate.
- **Marginal lending facility rate**, serves as the upper boundary:
 - Banks can borrow overnight funds from the ECB.
 - Higher rate discourages reliance on central bank borrowing for liquidity needs.

Evolution of Main Refinancing Rate

Figure: Evolution of Main Refinancing Rate (ECB) between oct-2019 to dec-2023



Economic Framework: Monetary Policy Impact

Impact of tightening (rate hikes):

- **Higher financing costs:** Banks face increased borrowing costs, reducing their willingness to lend.
- **Liquidity reduction:** Higher rates incentivize banks to retain reserves rather than lending.
- **Economic effects:** Decreased credit availability can slow down investments and consumption.

Impact of Loosening (Rate Cuts):

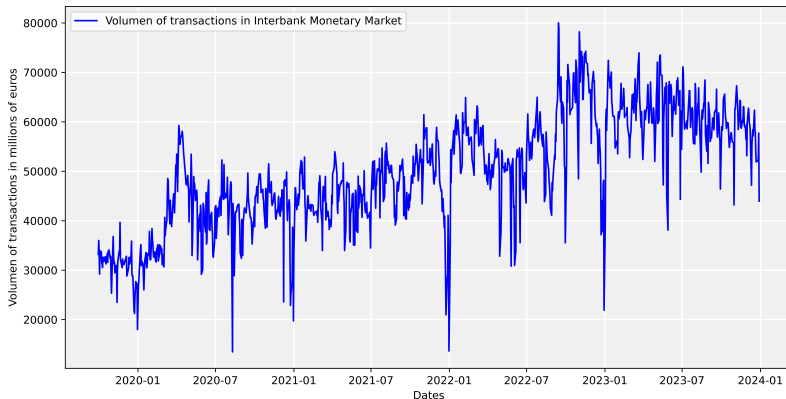
- **Stimulated lending:** Lower rates make borrowing cheaper, encouraging credit expansion.
- **Increased liquidity:** Banks are less incentivized to deposit reserves with the ECB, boosting interbank transactions.
- **Economic growth:** Facilitates investments and consumer spending.

Transmission to interbank market:

- Tighter policies reduce interbank transaction volumes due to higher costs.
- Loosening fosters interbank activity, stabilizing short-term funding needs.

Evolution of interbank volume transactions

Figure: Evolution of volume transactions (IMM) between oct-2019 to dec-2023



Descriptive statistics of IMM (1/2)

Table: Descriptive stats european interbank monetary market (millions euros)

Mean	Standard Deviation	Minimum	Maximum	Median
49801,51	11352,27	13472,00	80024,00	49697,50

Key metrics (October 2019 - December 2023):

- ❑ **Mean transaction volume:** High average liquidity in the interbank market.
- ❑ **Standard deviation:** Elevated volatility reflecting market uncertainty.
- ❑ **Minimum:** Periods of low liquidity, often during monetary tightening.
- ❑ **Maximum:** Peaks driven by high demand for liquidity during crises.
- ❑ **Median:** Consistent transaction volumes around the central value.

Descriptive statistics of IMM (2/2)

Notable trends:

① Phase 1 (2019-2022):

- Steady increase in transaction volumes (80% growth), supported by accommodative policies.
- Significant volatility post-2021 due to macroeconomic shocks (e.g., COVID-19).

② Phase 2 (2022-2023):

- Decline in transaction volumes following ECB's monetary policy tightening.
- Increased financing costs led to reduced liquidity in the market.
- Reflects sensitivity to macroeconomic shocks and monetary policy adjustments.

Implications:

- High volatility indicates sensitivity to monetary policy shifts.
- Liquidity constraints are more pronounced during rate hike cycles.

Methodology : Justification of Model Selection (1/2)

Why use ARDL-ECM, NARDL, and Koyck Models?

① Dynamic relationships:

- The European interbank market reacts over time to changes in the refinancing rate. ARDL captures both short- and long-term effects, while ECM provides insights into the speed of adjustment to equilibrium.

② Asymmetric reactions:

- Banks may react differently to increases and decreases in the refinancing rate. NARDL allows us to decompose and model these asymmetric effects explicitly.

③ Lagged effects of monetary policy:

- Changes in interest rates do not impact the interbank market instantaneously. Koyck transformation simplifies the representation of infinite lags, capturing gradual adjustment dynamics.

④ Handling non-stationarity:

- ARDL and ECM handle a mix of $I(0)$ and $I(1)$ variables, common in financial time series.

Relevance to this study:

- ① Investigating how monetary shocks propagate through the interbank market requires models capable of capturing the transmission mechanisms of policy changes. Nonlinear behaviors, such as differing responses to tightening vs. loosening policies.
- ② These models provide a robust framework to analyze transaction volumes and liquidity dynamics from October 2019 to December 2023.

Methodology: Theoretical Framework linear models

→ **ARDL(h, p) model:**

$$y_t = \mu + \sum_{i=1}^h \delta_i x_{t-i} + \sum_{j=1}^p \phi_j y_{t-j} + \varepsilon_t$$

where Y_t and X_t are $I(0)$.

→ **Koyck transformation:** simplifies infinite lags assuming geometric decay for measuring temporal impact:

$$y_t = \lambda y_{t-1} + (1 - \lambda)\mu + \delta_0 y_t + \varepsilon_t - \lambda \varepsilon_{t-1}$$

where λ measures the persistence of shocks.

→ **ARDL-ECM(h, p):**

$$\Delta y_t = \rho \xi_{t-1} + \mu + \sum_{i=1}^h \delta_i \Delta x_{t-i} + \sum_{j=1}^p \phi_j \Delta y_{t-j} + \varepsilon_t$$

where Δx_t and Δy_t are $I(0)$ and ξ_{t-1} long term relation $I(1)$.

Methodology: Theoretical Framework nonlinear models (1/3)

We can define the long run regression as¹ :

$$y_t = \beta^+ x_t^+ + \beta^- x_t^- + \varepsilon_t \quad (1)$$

where, x_t^+ et x_t^- are partials sum processes of positive and negative changes :

$$x_t^+ = \sum_{j=1}^t \Delta x_j^+ = \sum_{j=1}^t \max(\Delta x_j; 0)$$

,

$$x_t^- = \sum_{j=1}^t \Delta x_j^- = \sum_{j=1}^t \min(\Delta x_j; 0)$$

where y_t and x_t are scalar $I(1)$ variables, and x_t is decomposed as $x_t = x_0 + x_t^+ + x_t^-$.

¹Shin et al., 2013, Modelling Asymmetric Cointegration and Dynamic Multipliers in a Nonlinear ARDL Framework.

Methodology: Theoretical Framework nonlinear models (2/3)

We can define, following the previous terms, Nonlinear ARDL(p, q, r) with m exogenous variables as :

$$y_t = \sum_{j=1}^p \phi_j y_{t-j} + \sum_{i=1}^m \left(\sum_{j=1}^q \theta_j^{+'} \mathbf{x}_{i,t-j}^{+} + \sum_{j=1}^r \theta_j^{-'} \mathbf{x}_{i,t-j}^{-} \right) + \varepsilon_t$$

- Once the model is estimated, a restriction test on the coefficients must be performed.
- The Wald test is used to check if the two asymmetric parameters are different from zero.

Methodology: Theoretical Framework nonlinear models (3/3)

→ **Short-term multiplier:** Measures immediate impact:

$$\lambda_{ST} = \delta_t$$

→ **Long-term multiplier:** Captures cumulative effects over time:

$$\lambda_{LT} = \frac{\sum_{i=1}^h \delta_i}{1 - \sum_{j=1}^p \phi_j}$$

→ **Asymmetric multipliers:**

$$m_h^+ = \sum_h \frac{\partial Y_{t+i}}{\partial X_t^+}, \quad m_h^- = \sum_h \frac{\partial Y_{t+i}}{\partial X_t^-}$$

Tools: Models estimated using robust methods (Newey-West adjustments).

Results: Key Findings about Koyck Model

- ➡ **Slow decay** Summary results Koyck: The persistence parameter ($\lambda = 0.7$) indicates that 70% of the shock effect persists each period, leading to a gradual adjustment.
- ➡ **Mean and median lag**: The median time for 50% of the adjustment to materialize is approximately 9 days, and the average time is the same value.
- ➡ **Long-term dynamics**: The effect of monetary policy shocks is not immediate but lasts over several periods, reflecting the slow transmission process in the interbank market.

Results: Key Findings about Koyck Model

Partial Adjustments:

- **Adjustment coefficient:** Banks only adjust 30% of their transactions to the new equilibrium each period, indicating conservative strategies.
- **Gradual liquidity shift:** Market participants prefer phased adjustments, minimizing exposure to sudden rate changes.

Anticipations adaptatives:

- **Lagged expectations:** Market responses are influenced by prior expectations, adjusting progressively as new information is assimilated.
- **Expectational errors:** Initial responses to rate hikes are often underestimated, leading to corrections in subsequent periods.

Results: Key Findings about Koyck ARDL-ECM

- **Equilibrium adjustment** Summary results ECM Koyck: The error correction term ($\rho = -0.15$) suggests that deviations from the long-term equilibrium are corrected at a rate of 15% per period.
- **Time to equilibrium:** The system takes approximately 6-7 days to return to equilibrium after a shock.

Results: Key Findings about Koyck NARDL

Asymmetry (NARDL):

- ➡ **Stronger Impact of Rate Hikes** Summary results asymmetric Koyck:
Positive changes in the refinancing rate have a greater effect on interbank transaction volumes ($\Delta REFI^+$) compared to rate cuts ($\Delta REFI^-$).
- ➡ **Multiplier Differences:** Positive rate changes result in a long-term multiplier of $\beta^+ = 0.18$, whereas negative changes result in $\beta^- = 0.17$.

Results: Key Findings about Structural Changes with CUSUM test (1/2)

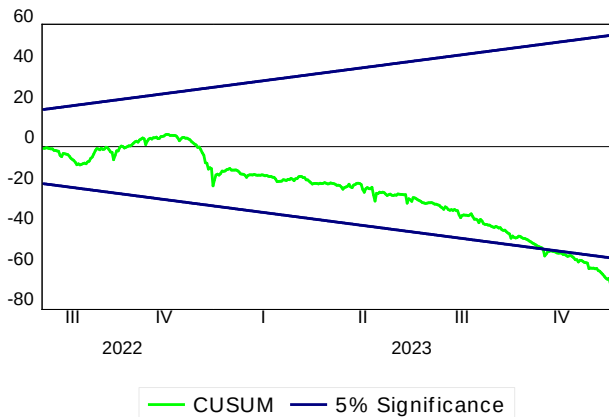


Figure: Caption

Results: Key Findings about Structural Changes with CUSUM test (2/2)

Structural changes:

- ➡ **2022 monetary policy shift:** The abrupt rate hikes by the ECB in late 2022 introduced a regime change, as confirmed by the CUSUM test.
- ➡ **Increased volatility:** Higher uncertainty in rate expectations led to amplified fluctuations in interbank transaction volumes.

Implications: Regulatory Insights (1/2)

Liquidity management:

- ❑ Gradual adjustments observed in the interbank market suggest the need for phased monetary policy interventions.
- ❑ Regulatory frameworks should focus on reducing interbank liquidity shocks through macroprudential tools.

Impact of asymmetry:

- ❑ Rate hikes disproportionately affect market liquidity compared to rate cuts.
- ❑ Policies should mitigate the amplified impact of tightening on small and medium-sized financial institutions.

Implications: Regulatory Insights (2/2)

Anticipations and communication:

- ❑ Adaptive expectations highlight the importance of clear, forward-looking communication by central banks.
- ❑ Transparent policy signaling can reduce market volatility and improve the efficiency of interbank adjustments.

Structural stability:

- ❑ Post-2022 volatility underscores the need for enhanced stress-testing of financial networks.
- ❑ Monitoring interconnectedness between institutions is critical to prevent contagion in times of rapid rate adjustments.

Results and Recommendations for Regulators

Key Results for Policy Design:

- ➔ **Inertia in adjustments:** Slow transmission of shocks (median lag of 9 days) calls for preemptive policy actions.
- ➔ **Asymmetric effects:** Regulatory focus is needed to address the outsized impact of rate hikes on liquidity constraints.
- ➔ **Structural breaks:** Regime changes (e.g., 2022) highlight the need for dynamic monitoring frameworks.

Recommendations:

- ➔ **Liquidity buffers:** Require banks to maintain sufficient liquidity reserves to withstand monetary tightening phases.
- ➔ **Enhanced interbank monitoring:** Develop real-time monitoring tools to track transaction volumes and volatility.
- ➔ **Stress testing:** Implement scenario-based stress testing to evaluate interbank resilience under rapid rate changes.
- ➔ **Macroprudential policy:** Consider countercyclical buffers to reduce the procyclicality of interbank lending during monetary tightening.

THANK YOU FOR YOUR
ATTENTION !

Results: Key Findings about Koyck Model

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Table: Summary of results Koyck Model

Variable	Coefficient	Prob.*
$LNTRANSACTION_{t-1}^{cvs,stat}$	0.687454	0.0000
$\Delta REFI^{cvs}$	0.176022	0.0028
ϕ_0	-0.000184	0.9590
R^2	0.476852	
Akaike info criterion	-1.436354	
Analyse des Résidus		
Homoscédasticité	Corrigé par Newey-West	
Autocorrélation	Corrigé par Newey-West	
Normalité	JB = 24124,68	0.0000

Results: Key Findings about Koyck ARDL-ECM

Retour

Table: Summary of results Koyck ARDL-ECM

Variable	Coefficient	Prob.*
ϕ_0	-0.000946	0.7825
$LNTRANSACTION S_{t-1}^{cvs,stat}$	0.767090	0.0000
$\Delta REFI_t^{stat}$	0.180208	0.0053
$\hat{z}_{t-1}$	-0.151855	0.0261
R^2	0.482611	
Akaike info criterion	-1.444712	
Analyse des Résidus		
Homoscédasticité	Corrigé par Newey-West	
Autocorrélation	Corrigé par Newey-West	
Normalité	JB = 23875,19	0.0000

Results: Key Findings about Koyck NARDL

Retour

Table: Summary of results Koyck NARDL

Variable	Coefficient	Prob.*
$LNTRANSACTION S_{t-1}^{cvs,stat}$	0.685650	0.0000
$\Delta REFI^{stat,-}$	0.174854	0.0031
$\Delta REFI^{stat,+}$	0.177626	0.0027
φ_0	0.003088	0.4939
R^2	0.477680	
Akaike info criterion	-1.435228	
Analyse des Résidus		
Homoscédasticité	Corrigé par Newey-West	
Autocorrélation	Corrigé par Newey-West	
Normalité	JB = 24515,10	0.0000